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NCALab: A Framework for Experimentation with Neural Cellular Automata

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Summary

Neural Cellular Automata (NCA) are lightweight iterative neural network models that can be employed in various image analysis tasks such as image segmentation, classification and generation. Initially proposed in 2020 (Mordvintsev et al., 2020), these models are recently getting attention thanks to their small size, their robustness and their overall versatility. In terms of accuracy, they are often on-par with state-of-the art models for the respective downstream task, while being orders of magnitude smaller in size. This is especially interesting in constrained settings, such as environments with limited compute resources. However, the training dynamics of NCAs are not yet fully understood, and there is potential for investigating practical tweaks to increase accuracy, reduce VRAM requirements and increase the overall training stability. `NCALab` provides a unified and extensible research framework for training and evaluating NCAs, conducting structured hyperparameter searches and prototyping applications that use NCAs for image analysis.

Neural Cellular Automata

Neural cellular automata (NCA) are a computational model combining the principles of Cellular Automata and Nneural Networks, allowing systems to learn and evolve over time. In NCAs, a grid of cells, each cell being capable of assuming a continuous, multi-dimensional state, changes based on a learned rule determined by neural networks rather than fixed algorithms. Typically, the inference is done in two steps, which are called *Perception* and *Update*. In the perception step, each cell's neighborhood is aggregated by applying multiple depth-wise convolutions. These convolutions can be learned, or hardcoded through static filters (e.g. Sobel or Laplace filters). In the *update* step, the residual cell update is computed by a multi-layer perceptron with a ReLU activation. This step is applied in a stochastic manner, meaning that only a portion of cells are activated in a single time step. The probability of the stochastic update is determined by a parameter called "fire rate", which is typically set to 0.5, activating only half of all cells in every time step. For more information, we want to point the reader to the comprehensive original paper on NCAs by Mordvintsev et al. (Mordvintsev et al., 2020).

State of the Field

NCAs are recently gaining attention especially in medical imaging, where they are deployed for various modalities in different downstream tasks, including 3D prostate segmentation on MRI (Kalkhof et al., 2023) (Kalkhof & Mukhopadhyay, 2023), image registration (Ranem et al., 2024) or image synthesis (Kalkhof et al., 2024), (Kalkhof et al., 2025). In most cases, they outperform other Convolutional Neural Network or Vision Transformer architectures in terms of *model size* and robustness (Kalkhof et al., 2023), while yielding similarly accurate predictions. Especially in constrained settings, such as radiology departments without access to GPU

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hardware, state-of-the art medical image segmentation is hardly possible with contemporary architectures based on Convolutional Neural Networks or Vision Transformers. NCAs, on the other hand, are a promising approach to enable AI support even in constrained settings.

Statement of Need

Research of NCA models and training has just started a few years ago, and there is no *unified* framework or reference implementation for training, evaluation and systematic experimentation with NCAs. Most NCA implementations live in custom codebases dedicated to the downstream task at hand, with little or now quality control or unit tests. Further, there is currently no set of best practices for designing NCA models with respect to their hyperparameters, such as the number of neurons, hidden channels or fire rate.

A systematic analysis of different NCA hyperparameters and architectural variations is difficult, as the research code of NCA contributions is typically organized in individual repositories with different frameworks and coding styles for each downstream task under investigation. Code bases often follow different approaches, even though the underlying backbone architecture is in most parts identical. In most cases, the NCA architecture can be defined by the number of input channels, hidden channels and output channels and the weights of the trained network. Since there is currently no unified framework, deployment of NCA models in practical applications (or as part of other learning pipelines) remains difficult.

`NCALab` provides a uniform and easy-to-use code base for various downstream tasks with NCAs as a packaged Python module. Within minutes, researchers and practitioners will be able to create prototypes for their ideas, inspired by the example tasks provided in this code repository. Code quality is maintained through automated type checking, unit testing (current code coverage: 66%) and linting.

We further provide weights for example tasks to enable post-hoc analyses on pre-trained NCA models. To our knowledge, post-hoc tasks like transfer learning or uncertainty estimation were not systematically evaluated for NCAs yet. `NCALab` enables such analyses, allowing to design uncertainty estimators tailored to NCA models and to evaluate them on multiple downstream tasks with little implementation overhead.

Features

`NCALab` provides dedicated models and example tasks for image analysis tasks, such as:

- Growing Neural Cellular Automata for emoji generation from a single pixel, akin to (Mordvintsev et al., 2020).
- Self-classifying MNIST digits, similar to the work of (Randazzo et al., 2020).
- Pixel-wise medical image segmentation of Endoscopic images on the Kvasir-SEG dataset (Jha et al., 2019).
- Multi-class medical image classification on subsets of the MedMNIST image dataset (Yang et al., 2023).

Until now, NCALab provides the following key features:

- Simplified creation, training and loading of NCA models for various image analysis tasks
- Streamlined grid search for model and training hyperparameters
- Tensorboard integration to monitor training progress
- k-fold cross validation
- Control over various hyperparameters of NCAs
- Finetuning by re-training the final layer of an NCA
- Visualization and animation of the NCA inference process
- Cascaded training of multi-scale NCAs for training on higher-resolution images

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- Uncertainty heatmap generation

Software Design

NCALab follows a mostly object-oriented design.

Specialized NCA models subclass the abstract class `AbstractNCAModel` in `ncalab.models`. For a bare-minimum implementation, subclasses should implement the `loss()` method, returning a `dict` of loss terms that can be visualized (the item with key `total` is backpropagated). Additionally, different hooks exist for modifying the inference process, e.g. for adjusting the prediction after unrolling all NCA steps in `post_prediction()`. Examples for specialized NCA models can be found in `ncalab.models.applications`.

Central to the inference process is the `Prediction` class, which stores the tensors processed by NCA models and allows to retrieve channel sections individually, e.g. through the `hidden_channels` property accessing the hidden channels. The interplay of abstract NCA model components and Predictions is illustrated in Figure 1.

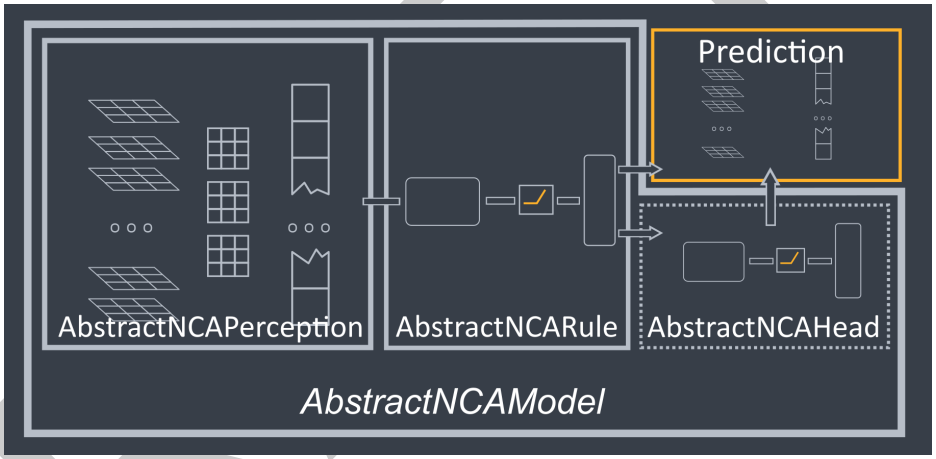


Figure 1: Illustration of abstract NCA model and its components, which produce Prediction objects.

NCAs are trained by passing them to a `BasicNCATrainer` object which stores training configuration and handles the entire training loop. The NCA trainer can be supplemented by hooks, which are subclasses of `TrainerHook`, e.g. for early stopping (`ncalab.training.earlystopping`).

Dependencies and Tooling

NCALab mostly builds on pytorch (Paszke et al., 2019), numpy (Harris et al., 2020) and matplotlib (Hunter, 2007). Code quality is backed by unit tests (pytest, coverage) and automated static code analysis through mypy and ruff. The project uses uv for fast and simplified dependency management. Code documentation is generated through Sphinx and is automatically uploaded to readthedocs. Releases of NCALab can be downloaded from the Python Package Index (pip).

Research Impact Statement

A conference paper utilizing NCALab was accepted for presentation in IPCAI 2025, and was published in the *International Journal of Computer-Assisted Radiology and Surgery* (Krumb & Mukhopadhyay, 2025). For creating this paper, NCALab was used to train models for image

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segmentation and monocular depth estimation. The trained models are exported to C headers and ported to a microcontroller.

AI usage disclosure

Generative AI has not been used in the scope of this project, neither for software creation nor for documentation or paper authoring.

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